

HELIOS 100 MW Solar PV EPC Project: Comprehensive Administration, Finance, and Regulatory Execution Report

1. Site Mobilisation (SUMMIT)

The mobilisation phase for a 100 MW utility-scale Solar PV Engineering, Procurement, and Construction (EPC) project establishes the operational and administrative foundation for the entire execution lifecycle. For the HELIOS project, situated in a remote region of Uzbekistan, mobilisation transcends the mere deployment of equipment; it requires the orchestration of a self-sustaining micro-city capable of supporting heavy civil, mechanical, and electrical works while adhering to international lender standards (IFC Performance Standards and EBRD Performance Requirements) and stringent Uzbek statutory regulations.

1.1 The Mobilisation Plan and Pre-Construction Prerequisites

A mobilisation plan for a 100 MW solar EPC project is a structured, 90-day roadmap designed to transition a greenfield site into a fully functional industrial construction zone. Before physical construction begins, several critical milestones must be achieved. The primary prerequisite is the acquisition of land rights and the initial construction permit from the Ministry of Construction (Davarchnazor)¹. Topographical surveys, hydrological studies, and geotechnical investigations must be validated to finalize the civil design. Concurrently, the site administration team must secure the site perimeter, establish temporary access roads, and deploy primary temporary facilities (TFC). An unexploded ordnance (UXO) sweep and comprehensive site clearance are mandatory prior to breaking ground. The mobilisation plan also includes the establishment of localized supply chains for bulk materials, such as concrete and aggregate, and the finalization of the site communications infrastructure.

1.2 Site Facilities Establishment

A 100 MW solar site requires extensive infrastructure to support a peak workforce ranging from 400 to 600 personnel. The mandatory facilities that must be established include modular, climate-controlled office cabins for the EPC contractor, the Owner's Engineer (OE), and client representatives, equipped with satellite or localized fiber internet to ensure seamless integration with corporate Enterprise Resource Planning (ERP) systems.

Accommodation camps must feature segregated blocks for expatriate engineers, third-country nationals (TCNs), and local workers, strictly adhering to IFC PS2 standards³. The canteen and messing facilities must feature industrial kitchens with deep cold storage, ensuring food safety compliance and accommodating diverse dietary and cultural requirements. A fully equipped site medical room is required, manned by a certified paramedic and a standby doctor, stocked with trauma kits, antivenom, and stabilization equipment for immediate emergency response.

Utilities form the backbone of the site. High-capacity diesel generators with N+1 redundancy are required for camp and office power, alongside reverse osmosis (RO) plants for potable

water and dedicated borewells or municipal tanker deliveries for construction water. Waste management facilities must include septic tanks for black and grey water, which are regularly evacuated by licensed vendors, and designated hazardous waste storage areas with secondary containment for generator oil and chemical storage.

1.3 Site Establishment Schedule

A typical site establishment schedule spans approximately eight to ten weeks, sequenced to ensure safety and logistical flow.

- Weeks one and two focus on the mobilisation of the early works contractor, completion of the site survey, unexploded ordnance (UXO) clearance, and the initial grading of the camp and office laydown areas.
- Weeks three and four involve the erection of perimeter fencing, the installation of site access gates and security watchtowers, and the delivery and leveling of modular office cabins.
- During weeks five and six, the hook-up of primary utilities occurs, including generators, water tanks, and septic systems, followed by the deployment of IT infrastructure and communications.
- Weeks seven and eight see the erection of the accommodation blocks and the main canteen, the mobilization of the site clinic and ambulance, and the execution of final safety inspections by the Owner's Engineer.
- By weeks nine and ten, the phased arrival of the core execution team begins, allowing for the commencement of piling and mechanical structural works.

1.4 Vehicle Fleet Requirements

A remote solar EPC site demands a robust, specialized fleet. Supervisory staff require 4x4 double-cab pick-up trucks to navigate ungraded tracks and adverse weather conditions. Heavy logistics dictate the need for flatbed trucks for module distribution across the array fields, telehandlers for unloading inverters and transformers, and specialized piling rigs. Site administration must manage a fleet of water bowsers, typically with a minimum 10,000-liter capacity, to ensure continuous dust suppression and supply for concrete batching plants. A dedicated, fully equipped 4x4 site ambulance is a non-negotiable requirement under occupational health and safety regulations to ensure rapid evacuation to the nearest regional hospital.

1.5 Workforce Mobilisation Process and Visa Requirements

Mobilising international engineers and supervisors to Uzbekistan involves navigating the regulatory framework of the Agency for External Labor Migration⁵. The process follows a strict sequence. First, the EPC contractor must conduct a labor market test to prove that the required specialized skills (such as high-voltage SCADA commissioning engineers) are unavailable within the local Uzbek labor pool⁷. Following this, the EPC obtains a corporate quota (Employer Permit) to hire foreign labor.

The individual worker must then obtain a Work Permit (Confirmation). Uzbekistan offers expedited processes for Highly Qualified Specialists—defined as individuals earning over \$60,000 annually and holding degrees from top universities—who benefit from a three-year permit process with a statutory fee of 440,000 UZS⁸. Standard foreign workers face higher

issuance fees and shorter permit durations⁷. Once the permit is secured, the worker applies for a Type E employment visa at the Uzbek embassy in their home country⁷. Upon arrival in Uzbekistan, the worker must be registered at the local migration office within three days and obtain a Personal Identification Number (PINFL) to operate legally⁷.

1.6 Accommodation Standards (IFC PS2)

Under IFC Performance Standard 2 (Labor and Working Conditions) and EBRD Performance Requirement 2, the EPC contractor must provide safe, sanitary, and non-restrictive housing. The practice of "hot bedding"—sharing beds across different working shifts—is strictly prohibited¹⁰. The net internal area must range from 10 to 12.5 cubic meters per person¹¹. Individual beds must measure at least 1.8 meters by 0.6 meters, with sufficient space between beds to allow comfortable access¹⁰.

Sanitary facilities require a ratio of at least one toilet and one shower per 10 to 15 individuals, featuring anti-slip flooring, adequate ventilation to prevent mold, and privacy locks¹⁰. The camp infrastructure must be engineered to withstand the extreme winter temperatures characteristic of Uzbekistan, ensuring that heating systems maintain comfortable living conditions and that water pipes are insulated against frost¹⁰. Furthermore, workers' freedom of movement to and from the employer-provided accommodation must not be unduly restricted, ensuring alignment with international human rights standards³.

2. HR Administration (SUMMIT)

The Human Resources function on a utility-scale construction site is a critical compliance and operational mechanism that protects the EPC contractor from statutory fines, labor disputes, and lender audits.

2.1 Maintenance of HR Records

Site HR must maintain an exhaustive and meticulously organized dossier for every individual entering the site. This includes biometric or RFID-backed timesheets, which are critical for validating the project cost ledger and substantiating progress. Verified copies of passports, visas, and PINFLs must be kept on file⁹. Health and safety compliance requires the retention of medical fitness-to-work certificates, signed induction records, and specialized training logs for high-risk activities such as working at heights or high-voltage switching. Under the 2023 Uzbekistan Labor Code, electronic employment contracts and rigorous working hour logs—enforcing a maximum 40-hour standard work week with strict overtime controls—are mandatory¹².

2.2 The Site Induction Process

No individual may enter the active construction zone without undergoing a formal site induction. Conducted primarily by the Health, Safety, and Environment (HSE) Department with administrative oversight from HR, the induction covers site-specific hazards, emergency muster points, and evacuation protocols. Critically, it covers the local grievance redress mechanism (GRM) mandated by the EBRD and IFC, ensuring that all workers understand how to report workplace or community grievances without fear of retaliation¹³. The induction also covers culturally sensitive behavior and zero-tolerance policies regarding alcohol consumption,

harassment, and ecological damage. Attendees must pass a comprehension test and sign an induction log, after which they receive a hardhat sticker and a site ID badge.

2.3 Individual Mobilisation and Demobilisation

Individual mobilisation requires HR to coordinate flight ticketing, airport transfers, camp room allocation, and the issuance of Personal Protective Equipment (PPE). The demobilisation process is equally rigid. Before departing the project, a worker must complete a comprehensive clearance checklist, returning all company assets, including laptops, radios, and specialized tools. HR processes the final financial settlement, cancels the work permit to free up the corporate labor quota, and executes the exit visa process, ensuring that the worker departs with no lingering tax liabilities in Uzbekistan.

2.4 Site Allowances Structure

To incentivize deployment to remote project locations, the compensation package for engineers and supervisors includes a base salary supplemented by a remote site allowance, often termed a hardship premium. Expatriate staff typically operate on a strict rotation schedule—such as eight weeks on-site followed by two weeks of paid leave—with international flights fully funded by the EPC contractor. Per diems are provided for transit travel days. On-site accommodations, laundry, and meals are provided in kind, meaning they are not disbursed as cash, which simplifies personal income tax calculations and ensures uniform living standards across the workforce.

2.5 Payroll Management on Site

Payroll processing integrates data from site timesheets, HR records (such as sick leave and annual leave), and finance. Under Uzbek statutory law, payroll calculations require the deduction of a 12% flat rate for Personal Income Tax (PIT) from the employee's gross salary, alongside a 12% Unified Social Tax paid by the employer¹². The finance team utilizes this gross payroll data to debit the respective Work Breakdown Structure (WBS) cost codes in the project ledger. This integration ensures that labor costs are accurately capitalized against the specific project asset being constructed, such as civil foundations or electrical stringing.

3. Project Finance & Cash Flow (LEDGER2)

3.1 The Project Cash Flow Statement

The project cash flow statement for an EPC project models the total liquidity required to execute the works over the project lifecycle. It is structured to track cash inflows—specifically Advance Payments, Interim Payment Certificates, and Retention Releases—against massive capital outflows, including the procurement of PV modules, inverters, civil subcontracts, payroll, and logistics. Because EPC contracts are highly capital-intensive and front-loaded with procurement costs, the cash flow statement serves to highlight peak negative cash positions, which dictate the size and timing of the contractor's working capital facility requirements.

3.2 Translating the S-Curve into a Cash Flow Forecast

The S-curve represents planned physical progress and earned value over time. However, physical progress does not equate to immediate liquidity. Translating the S-curve into a cash flow forecast requires the application of commercial lag factors. For example, if the S-curve

dictates that 10% of physical progress is achieved in Month 3, the contractor submits the invoice at the end of Month 3. This invoice is certified by the Owner's Engineer by mid-Month 4 and paid by the Employer by the end of Month 4, assuming standard 56-day FIDIC payment terms¹⁵. Consequently, physical execution and cost incurrence outpace cash receipt, creating a structural cash deficit that the Finance Controller must actively manage.

3.3 The Payment Application Process (FIDIC)

Under FIDIC Yellow or Silver book conditions, the Contractor submits an Application for an Interim Payment Certificate (IPC) at the end of each month¹⁵. This application aggregates the value of executed works, materials delivered to the site but not yet installed (under Sub-Clause 14.5), and approved variations¹⁵. The Owner's Engineer evaluates the application, makes any necessary deductions for defective work or lack of substantiation, and issues the IPC to the Employer¹⁵. The application automatically deducts the agreed advance payment recovery percentage and the contractual retention money percentage¹⁵.

3.4 Bank Guarantees in EPC

Bank guarantees are unconditional financial instruments securing the Employer against Contractor default. They are tracked rigorously by the finance team.

- **Advance Payment Guarantee (APG):** Issued for 10% to 20% of the contract value. It amortizes (reduces in value) as the advance is recovered through deductions in successive interim payments.
- **Performance Bond / Guarantee (PBG):** Typically 10% of the contract price. This instrument remains valid until the issuance of the Taking-Over Certificate (TOC) or the end of the Defects Notification Period (DNP), serving as security for the execution of the works¹⁷.
- **Retention Money Guarantee (RMG):** Used when the Contractor wishes to receive the cash that would otherwise be held as retention. The RMG replaces the cash holdback, freeing up critical working capital for the contractor¹⁷.

3.5 Letters of Credit (LC) for Equipment Procurement

Procuring Tier-1 solar modules involves transactions worth tens of millions of dollars. The EPC utilizes Letters of Credit (LCs) to secure this global supply chain. The EPC's bank (Issuing Bank) guarantees payment to the Supplier's bank (Advising Bank) upon the presentation of compliant shipping documents, such as the Bill of Lading, Certificate of Origin, and Commercial Invoice, in strict accordance with UCP 600 rules. The Finance Controller tracks LC issuance dates, margin deposits, amendment fees, and maturity dates, ensuring that equipment delivery aligns with customs clearance and site installation schedules.

3.6 Retention Money

Retention money is a percentage—usually 5% to 10%—deducted from every IPC to incentivize the Contractor to complete the works and rectify any defects¹⁷. It accumulates up to a predefined limit, commonly 5% of the total contract price¹⁸. Under FIDIC Sub-Clause 14.9, it is released in two tranches: the first 50% upon the issuance of the Taking-Over Certificate (TOC) for the works, and the remaining 50% upon the expiry of the Defects Notification Period (DNP) and the subsequent issuance of the Performance Certificate¹⁸.

3.7 Liquidity Risk Triggers

The Finance Controller flags a liquidity risk when specific operational or commercial triggers occur. These include:

1. **Under-certification:** The Owner's Engineer certifies significantly less than the applied IPC value, restricting anticipated cash inflow¹⁹.
2. **Payment Delays:** The Employer breaches the contractual 56-day payment window¹⁵.
3. **Currency Fluctuations:** Mismatches between the currency of the EPC contract (e.g., USD) and local operational costs (e.g., UZS) leading to unhedged FX losses.
4. **Customs Demurrage:** Delays in clearing solar modules resulting in heavy storage fees and the delay of crucial IPC milestones.

3.8 The Cost Ledger and Budget Codes

The cost ledger is a highly granular accounting database tracking every incurred project cost against the approved baseline budget. It utilizes a standardized Work Breakdown Structure (WBS) or cost code system. By comparing committed costs (Purchase Orders issued) and actual costs (invoices paid) against the budget codes, the Finance Controller conducts Earned Value Management (EVM) to forecast the Estimate at Completion (EAC) and identify cost overruns in real-time.

4. Statutory Permits & Regulatory Approvals (PERMIT)

Executing a 100 MW solar project in Uzbekistan requires navigating a complex matrix of regulatory authorities. The failure to secure a single permit can paralyze project execution.

4.1 Exhaustive List of Required Permits and Approvals

1. Foreign Enterprise Registration

- **Issuing Authority:** State Services Center (DXM) via my.gov.uz²⁰
- **Typical Timeline:** 1 to 3 working days.
- **Submission Requirements:** Company charter, founder passports, apostilled parent company documents, PINFL of the designated director, and proof of registered legal address²⁰.
- **Project Phase:** Pre-Mobilisation.

2. Land Allocation and Lease Agreement

- **Issuing Authority:** Cabinet of Ministers / Local Regional Hokimiyat²²
- **Typical Timeline:** 30 to 90 days.
- **Submission Requirements:** Presidential decree supporting the PPP/RES project, detailed topographical surveys, cadastral boundary coordinates, and agricultural land conversion justifications²².
- **Project Phase:** Pre-Mobilisation.

3. State Environmental Expertise (ZVER / ZEP)

- **Issuing Authority:** Center for State Ecological Expertise (Ministry of Ecology, Environmental Protection and Climate Change)²⁴
- **Typical Timeline:** 30 to 45 days.
- **Submission Requirements:** Draft Environmental Impact Assessment (EIA) materials,

official minutes from public hearings with local communities, cumulative impact assessments, and a detailed mitigation plan²⁵.

- **Project Phase:** Pre-Construction. (Note: Banks cannot finance projects lacking this positive conclusion)²⁷.

4. Construction Permit

- **Issuing Authority:** Inspectorate for Control in the Sphere of Construction and Housing (Davarchnazor)¹
- **Typical Timeline:** 15 to 30 days.
- **Submission Requirements:** Approved detailed design documentation, positive ZVER environmental conclusion, contractor licenses, and proof of technical supervision appointments².
- **Project Phase:** Pre-Construction.

5. Grid Connection Agreement & Technical Conditions (TU)

- **Issuing Authority:** National Electric Grid of Uzbekistan (NEGU)²⁸
- **Typical Timeline:** 30 to 60 days.
- **Submission Requirements:** Single line diagrams, load flow and short-circuit studies, generation profiles, and substation tie-in designs²⁸. All documents must be in Uzbek or notarized translations.
- **Project Phase:** Design and Procurement.

6. Electrical Installation Approval (Energisation Consent)

- **Issuing Authority:** Uzenergoinspeksiya³⁰
- **Typical Timeline:** 15 to 30 days.
- **Submission Requirements:** As-built drawings, approved relay setting calculations, Site Acceptance Test (SAT) reports, primary injection testing results, and earthing system validation certificates³¹.
- **Project Phase:** Commissioning (Pre-Energization).

7. Import Permits and Customs Clearance

- **Issuing Authority:** State Customs Committee³²
- **Typical Timeline:** 3 to 7 days per shipment.
- **Submission Requirements:** E-Contract registration via the Unified System, Commercial Invoice, Bill of Lading, Certificate of Origin, and strictly mandated proof that the solar equipment is listed on the BNEF Tier-1 registry³².
- **Project Phase:** Execution / Procurement.

8. Labour Quota and Work Permits

- **Issuing Authority:** Agency for External Labor Migration⁵
- **Typical Timeline:** 30 to 60 days for corporate quotas; 15 to 30 days for individual confirmations.
- **Submission Requirements:** Labor market test documentation proving lack of local skills, corporate registration documents, worker passports, notarized diplomas, and medical certificates⁷.
- **Project Phase:** Mobilisation.

9. Fire Inspection Approval

- **Issuing Authority:** Ministry of Emergency Situations (MES)³¹
- **Typical Timeline:** 15 to 30 days.
- **Submission Requirements:** Fire alarm system layouts, fire extinguisher placement plans, civil defense approval, and MSDS (Material Safety Data Sheets) for battery/transformer rooms³¹.
- **Project Phase:** Commissioning.

10. Final Commissioning / Operating Licence

- **Issuing Authority:** State Acceptance Commission (Multi-agency committee)
- **Typical Timeline:** 30 to 60 days.
- **Submission Requirements:** Consolidation of all prior permits, performance ratio (PR) test results, Taking-Over Certificate (TOC) from the Employer, and finalized As-Built documentation.
- **Project Phase:** Close-out.

5. In-Country Operational Requirements (Uzbekistan Specific)

5.1 Banking Requirements and Currency Controls

Foreign contractors operating in Uzbekistan must navigate a highly regulated banking environment. To execute cross-border transactions and pay overseas suppliers, all international contracts must be registered in the Unified Electronic Information System of Foreign Trade Operations (UEISFTO), commonly referred to as the E-Contract system (contract.customs.uz)³³. This system is jointly monitored by the Central Bank and the State Customs Committee. Commercial banks are legally prohibited from transferring funds from local accounts to overseas accounts unless the underlying contract is registered in this portal³⁷. While the repatriation of legitimate profits is permitted, it requires rigorous tax clearance and documentary evidence of achieved project milestones.

5.2 Tax Obligations (Permanent Establishment)

Under Article 36 of the Tax Code of Uzbekistan, a foreign EPC contractor creates a Permanent Establishment (PE) if its construction, assembly, or supervisory activities continue for a period exceeding 183 days (or 6 months, depending on specific Double Taxation Treaties) within any 12-month period³⁸. Once a PE is formed, the contractor is subject to a 15% Corporate Income Tax (CIT) on the net profit attributable to the PE¹⁴. Furthermore, the PE must register as a Value Added Tax (VAT) payer, subject to the standard 12% rate¹⁴. If the EPC utilizes offshore sub-consultants for design or engineering, those service payments are subject to a 20% Withholding Tax (WHT). However, this WHT can be reduced or eliminated if a Double Taxation Treaty (DTT) exists between Uzbekistan and the subcontractor's domicile, provided a valid tax residency certificate is presented¹⁴.

5.3 Local Content and Import Duty Exemptions

Uzbekistan is aggressively promoting its renewable energy sector while simultaneously attempting to foster local manufacturing. While the standard 12% VAT applies to imports, technological equipment not produced locally is frequently exempt from customs duties under

specific investment agreements¹⁴. However, a critical regulatory shift mandates that starting January 1, 2025, only solar panels, inverters, and battery storage systems that are listed on the BloombergNEF (BNEF) Tier-1 list are permitted to be imported into the country³⁴. The Customs Committee checks this list dynamically; if a manufacturer drops off the Tier-1 list in a given quarter, their equipment will be denied customs clearance, representing a severe supply chain risk³⁴.

5.4 Corporate Registration Requirements

To operate legally, foreign contractors typically establish a Foreign Enterprise Limited Liability Company (FE LLC) or register a Branch Office. The registration process has been heavily digitized via the State Services Center (my.gov.uz) and the "Birdarcha" (One Stop Shop) portal⁹. While historical regulations required a statutory minimum capital of 400 million UZS for an FE LLC, recent investment climate reforms have largely eliminated strict capital minimums for standard LLC structures, allowing for rapid incorporation²⁰. Corporate directors must obtain a Personal Identification Number (PINFL) to open bank accounts and execute legally binding documents²¹.

6. Insurance Management (LEDGE / LEDGER2)

Risk transfer via insurance is a strict lender condition (IFC/EBRD) and a statutory requirement under Uzbek law for construction projects.

6.1 Required Insurance Policies

- **Contractor's All Risk (CAR) / Erection All Risk (EAR) Policy:** Mandatory under Cabinet of Ministers Resolution No. 532 for projects involving state funds or guarantees⁴⁷. The CAR policy covers physical damage to the permanent works (solar arrays, substations) resulting from natural disasters, fire, or accidental damage during execution. The sum insured must equal the full contract value. Typical exclusions include willful negligence, war, acts of terrorism, and normal wear-and-tear.
- **Third Party Liability (TPL) Policy:** Frequently bundled with the CAR policy, TPL covers the contractor's legal liability for bodily injury or property damage to third parties arising directly from construction activities⁴⁸.
- **Marine Cargo Insurance:** Covers PV modules, transformers, and inverters during international transit from the factory to the project site, usually written on a comprehensive "warehouse-to-warehouse" basis.
- **Workmen's Compensation / Employer's Liability:** Mandatory under Uzbek Law (ZRU-210)⁴⁷. It provides financial protection for employees suffering occupational injuries, diseases, or fatalities.
- **Professional Indemnity (PI):** Required if the EPC is performing the engineering design, covering financial losses arising from design defects or professional negligence.

6.2 The Insurance Claim Process

In the event of a loss event (for example, a severe windstorm damaging installed but un-commissioned solar panels), the claim process must be executed rapidly to secure liquidity for replacement materials:

1. **Immediate Notification:** The insurer must be notified in writing within a strict statutory timeframe, often 7 calendar days in Uzbekistan, to avoid claim invalidation⁴⁸.
2. **Mitigation:** The contractor is legally obligated to take immediate, reasonable steps to mitigate further loss (e.g., securing loose panels or erecting temporary weather protection).
3. **Surveyor Appointment:** The insurance company appoints an independent loss adjuster to inspect the site, determine the root cause, and review the extent of the damage.
4. **Documentation:** The contractor submits comprehensive substantiation, including Bills of Materials (BOQs) for the damaged elements, replacement invoices, weather reports, and internal incident logs.
5. **Settlement:** Under Uzbek regulations governing mandatory construction insurance, uncontested claims must be paid within 5 banking days of the finalized damage assessment report being signed by the commission⁴⁸.

6.3 The Insurance Register

The Insurance Register is maintained by the Finance Controller to track policy validity and claims. It tracks the Policy Type, Insurer, Policy Number, Sum Insured, Deductible, Expiry Date, and Open Claims Status (see Section 7.7 for template).

7. Real Document Templates — Full Detail

7.1 Site Establishment Checklist

#	Category	Item Description	Status	Target Date	Responsibility
1	Permits	Initial Construction Permit obtained from Davarchnaz or	Pending	15-Mar-27	PERMIT
2	Civil	Camp grading, leveling, and perimeter fencing completed	Complete	01-Apr-27	Construction Mgr

3	Utilities	2x 500kVA Prime Diesel Generators installed and load tested	Pending	10-Apr-27	MEP Manager
4	Utilities	Borewell drilled or 10kL water tanks installed	Complete	05-Apr-27	Logistics
5	HSE	Site Clinic established with paramedic & fully equipped ambulance	Pending	12-Apr-27	HSE Manager
6	IT	VSAT / Fiber internet active in primary engineering office	Pending	15-Apr-27	IT Admin
7	HR	Camp accommodation ready (12.5m3 per person verified)	Pending	20-Apr-27	SUMMIT
8	Waste	Septic tanks installed and hazardous	Pending	22-Apr-27	HSE Manager

		waste containmen t built			
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7.2 Project Cash Flow Statement (18-Month EPC, USD '000)

Months 1 - 9

Desc ription	M1	M2	M3	M4	M5	M6	M7	M8	M9
Inflo ws									
Adva nce Pmt (10%)	10,0 00	0	0	0	0	0	0	0	0
IPC Paym ents	0	0	0	2,50 0	4,20 0	8,100	12,00 0	15,00 0	14,00 0
Total Inflo w	10,0 00	0	0	2,50 0	4,20 0	8,10 0	12,0 00	15,0 00	14,0 00
Outfl ows									
Proc urem ent (Mod ules)	(2,00 0)	(5,00 0)	(8,00 0)	(2,00 0)	(1,00 0)	(500)	(500)	(500)	0
Civil Subc ontra	(500)	(1,20 0)	(1,50 0)	(2,00 0)	(2,50 0)	(3,00 0)	(3,00 0)	(2,50 0)	(1,00 0)

cts									
Payroll & Admin	(200)	(250)	(300)	(350)	(400)	(450)	(500)	(500)	(450)
Total Outflow	(2,700)	(6,450)	(9,800)	(4,350)	(3,900)	(3,950)	(4,000)	(3,500)	(1,450)
Net Cash Flow	7,300	(6,450)	(9,800)	(1,850)	300	4,150	8,000	11,500	12,550
Cumulative Cash	7,300	850	(8,950)	(10,800)	(10,500)	(6,350)	1,650	13,150	25,700

Months 10 - 18

Description	M10	M11	M12	M13	M14	M15	M16	M17	M18
Inflows									
IPC Payments	10,000	6,000	3,000	1,000	500	0	0	0	0
Retention Release 1	0	0	0	0	2,500	0	0	0	0
Total	10,0	6,00	3,00	1,00	3,00	0	0	0	0

Inflow	00	0	0	0	0				
Outflows									
Electrical Works	(3,000)	(2,000)	(1,000)	(500)	(200)	0	0	0	0
Commissioning	(500)	(500)	(400)	(200)	(100)	0	0	0	0
Payroll & Admin	(400)	(350)	(300)	(200)	(100)	(50)	(50)	(50)	(50)
Total Outflow	(3,900)	(2,850)	(1,700)	(900)	(400)	(50)	(50)	(50)	(50)
Net Cash Flow	6,100	3,150	1,300	100	2,600	(50)	(50)	(50)	(50)
Cumulative Cash	31,800	34,950	36,250	36,350	38,950	38,900	38,850	38,800	38,750

7.3 Bank Guarantee Tracker

BG Ref No.	BG Type	Issuing Bank	Beneficiary	Value (USD)	Issue Date	Expiry Date	Status	Amortized Bal.
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APG-001	Advance Pmt	KDB Bank Uz	NEGU	10,000,000	01-Jan-27	31-Dec-27	Active	6,500,000
PBG-002	Perf. Bond	NBU	NEGU	10,000,000	01-Jan-27	30-Jun-28	Active	10,000,000
RMG-003	Ret. Money	KDB Bank Uz	NEGU	2,500,000	15-Oct-27	30-Jun-29	Draft	N/A

7.4 Permit Register

Permit Name	Issuing Authority	Submission Date	Target Approval	Actual Approval	Status	Owner
ZVER (State Eco Expertise)	Min of Ecology	10-Jan-27	25-Feb-27	22-Feb-27	Approved	PERMIT
Foreign Labor Quota	Agency Ext Labor	15-Jan-27	15-Mar-27	10-Mar-27	Approved	SUMMIT
Construction Permit	Davarchnazor	01-Mar-27	30-Mar-27	-	Pending	PERMIT
E-Contract Registration	Customs / Central Bank	05-Mar-27	07-Mar-27	06-Mar-27	Approved	LEDGER2
Grid Connecti	NEGU	10-Apr-27	10-Jun-27	-	Prep	Engineering

on (TU)						
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7.5 Site Induction Record

Project: HELIOS 100 MW Solar PV | **Date:** 12-May-27 | **Instructor:** A. Ivanov (HSE)

Employee Name	Passport / ID	Company / Sub	Topics Covered	Test Score	Signature
John Smith	US-1234567	EPC Main	UXO, Heat Stress, GRM, IFC limits	100%	[Signed]
Timur Aliev	UZ-9876543	Civil Sub	UXO, Heat Stress, GRM, IFC limits	95%	[Signed]
Maria Garcia	ES-6543210	QA/QC Consult.	UXO, Heat Stress, GRM, IFC limits	100%	[Signed]

7.6 Worker Mobilisation Record

Name	Role	Visa Type	Work Permit No.	Flight Arrival	Camp Room	Med. Clear Date	Induction Status
D. Martinez	SCADA Eng	Type E	WP-27-0012	14-May-27	Block A-12	01-May-27	Scheduled
S. Kumar	QA/QC Mgr	Type E	WP-27-0015	16-May-27	Block B-04	03-May-27	Scheduled
A. Petrov	Civil Supv.	Type E	WP-27-0018	10-May-27	Block A-08	28-Apr-27	Completed

7.7 Insurance Register

Policy Type	Insurer	Policy No.	Sum Insured (USD)	Deductible	Expiry Date	Claim Status	Open Value
CAR / EAR	Gross Insurance	CAR-27-99	100,000,000	50,000	31-Dec-28	Nil Claims	0
Workmen's Comp	Ishonch Sug'urt a	WC-27-14	Statutory	Nil	31-Dec-27	1 Open	Pending
Marine Cargo	Gross Insurance	MC-27-88	60,000,000	10,000	31-Dec-27	Nil Claims	0
TPL	Gross Insurance	TPL-27-10	10,000,000	5,000	31-Dec-28	Nil Claims	0

7.8 Cost Code Structure (WBS)

Cost Code	Description	Sub-Code Example
1000	Project Preliminaries & Admin	1100: Staff Payroll, 1200: Camp Ops, 1300: Visas/Permits, 1400: Vehicles
2000	Civil Works	2100: Topography/UXO, 2200: Grading/Roads, 2300: Piling, 2400: Fencing
3000	Mechanical Works	3100: Tracker/Structure Assembly, 3200: PV Module Mounting, 3300: Torqueing

4000	Electrical Works (DC/AC)	4100: String Cabling, 4200: Inverter Install, 4300: 33kV Trenching/Pulling
5000	Substation & Grid (220kV)	5100: Transformer Install, 5200: GIS/AIS Switchgear, 5300: OHL Tie-in
6000	Procurement (Materials)	6100: PV Modules, 6200: Inverters, 6300: Cables, 6400: Steel Structures
7000	Logistics & Freight	7100: Ocean Freight, 7200: Inland Transport, 7300: Customs Duties/Fees
8000	Testing & Commissioning	8100: Cold Commissioning, 8200: Hot Commissioning, 8300: Grid Code Tests

8. Admin/Finance Interfaces

The seamless execution of the HELIOS project relies entirely on the interconnected data flows between administrative, financial, and operational departments. Siloed information inevitably leads to project delays and cost overruns.

- **With Supply Chain Management (SCM):** The SCM team cannot release purchase orders for imported equipment without Finance registering the transaction in the E-Contract system to comply with Central Bank currency controls³³. Finance (LEDGER2) monitors LC margin requirements to optimize treasury cash balances, while SCM tracks manufacturing milestones to ensure LC drawdowns align with physical delivery.
- **With Contracts:** The Contracts department defends IPC valuations against arbitrary reductions by the Owner's Engineer. Finance relies on Contracts to manage the validity of Bank Guarantees. If project delays push completion past the current PBG expiry date, Contracts must coordinate an extension well in advance to avoid a hostile "extend or pay" call from the Employer.
- **With Planning:** The Planning department generates the physical S-curve. Finance takes this baseline schedule and translates it into the cash flow forecast by applying commercial realities: contractual payment lags, advance payment amortizations, and retention deductions. A delay in the planner's P6 schedule must immediately reflect as a deferred cash inflow in the finance models.

- **With HR/Admin:** HR (SUMMIT) controls the biometric timesheets, sick leave, and rotation schedules. Finance processes this raw data to execute the payroll, withholding the mandatory 12% PIT and 12% Social Tax¹², and subsequently feeds the cost ledger to track labor efficiency (actual man-hours versus budgeted man-hours).
- **With Statutory Bodies:** The statutory liaison (PERMIT) translates highly technical engineering outputs—such as single-line diagrams, relay settings, and short-circuit calculations—into the specific administrative formats required by NEGU and Uzenergoinspeksiya to secure grid connection and energization consents²⁸.

9. Common Admin, Finance, and Permit Problems

9.1 Foreign Worker's Work Permit is Delayed

Risk: Critical path resources, such as specialized high-voltage testing engineers, cannot legally deploy, directly threatening the commissioning schedule. **Contingency:** Under no circumstances should the individual enter Uzbekistan on a tourist visa to perform manual labor; doing so triggers severe fines, project suspension, and deportation under Uzbek law⁷. The immediate contingency is to apply for a short-term business visa⁵⁰. Under a business visa, the engineer may legally attend the site for inspections, meetings, and supervision, but physical execution must be delegated to locally permitted staff under their strict verbal guidance. Simultaneously, the SUMMIT coordinator escalates the work permit blockage to the Ministry of Investment, Industry, and Trade (MIIT) to request expedited processing based on the project's strategic national importance.

9.2 Client Delays Payment Certification

Risk: The Employer's Engineer systematically under-certifies the IPC or breaches the 56-day payment window (FIDIC Sub-Clause 14.7), triggering a severe liquidity crisis for the EPC contractor¹⁵. **Contingency:** LEDGER2 must treat under-certification immediately as a valuation dispute, compiling irrefutable substantiation such as signed delivery notes and joint measurement sheets¹⁹. If the payment deadline passes, LEDGER2 instructs Contracts to issue a formal Notice of Delayed Payment. Under FIDIC Sub-Clause 14.8, the EPC claims financing charges (interest) on the delayed amount¹⁵. If non-payment persists, the ultimate contractual escalation is a 21-day Notice of Intention to Suspend Works under FIDIC Sub-Clause 16.1, leveraging schedule impacts to force Employer compliance¹⁵.

9.3 Customs Authority Delays Equipment Clearance

Risk: High-value solar modules are held at the border, incurring massive daily demurrage charges and halting array installation. **Contingency:** In Uzbekistan, a primary trigger for solar customs delays is the failure to prove compliance with the BNEF Tier-1 mandate³⁴. PERMIT and SCM must preemptively secure official BNEF listing documentation valid specifically *for the quarter of importation*³⁴. If cargo is held, PERMIT liaises directly with the State Customs Committee, ensuring the E-Contract data aligns perfectly with the commercial invoice, packing list, and Certificate of Origin³². The contractor formally requests "green channel" expediting, emphasizing the risk to the national energy grid timeline.

9.4 Environmental Inspection Finds Site Non-Compliant

Risk: An audit by the Ministry of Ecology or the Lender's Independent E&S Consultant (LESC) discovers improper waste disposal or deviations from the ZVER/ESIA conditions (e.g., hazardous runoff)¹³. **Contingency:** The inspector issues a non-conformance report. PERMIT and HSE must immediately halt the offending activity. A Corrective Action Plan (CAP) is drafted within 48 hours, outlining immediate remediation (e.g., excavating contaminated soil, deploying secondary containment) and long-term preventative measures (e.g., retraining staff). The CAP is submitted to the Lenders (per the Environmental and Social Action Plan - ESAP mechanism) and local Uzbek authorities¹³. Failure to act swiftly risks the immediate suspension of the Construction Permit by Davarchnazor.

9.5 Bank Guarantee Expiry Approaching but Extension Not Approved

Risk: The project is delayed due to force majeure or Employer risk events, but an Extension of Time (EOT) claim remains pending. The Performance Bond (PBG) is set to expire in 30 days.

Contingency: Employers will invariably issue a hostile "extend or pay" demand to the issuing bank if a BG is allowed to lapse, immediately seizing the funds¹⁷. LEDGER2 tracks BGs rigidly with an automated 60-day early warning alert. Upon the alert, LEDGER2 instructs the issuing bank to extend the validity of the PBG to cover the projected delay period, regardless of the EOT status. The financial cost of this BG extension is captured in the cost ledger and subsequently added to the Contractor's EOT prolongation cost claim to be recovered from the Employer.

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